

2) Geometric distribution

Success probability =  $p$   $E[X] = 1/p$   
 fail =  $(1-p)$

expectation that  
 someone passes by and  
 joins the group

$E[X] = \frac{1}{p} + p(1) + (1-p)(1+E)$  Law of total probability;

$$E = \frac{1}{p} + (1-p)E$$

$\frac{1}{p}$  = find the first person,

$$E - (1-p)E = \frac{1}{p} + 1$$

$p(1)$  = find 1 more person (trio),

$$pE = 1 + p$$

$(1-p)$  = Failure,  $(1+E)$  = start over,

$$E = \frac{1+p}{p}$$

when failure there is a new  
 Expectation. Then solve for  $E$

Every  $\lambda$  minutes, new person. Multiply Every Expectation  
 by  $\lambda$  to get:

$$E[T] = \frac{1+p}{p} \cdot \lambda$$

$$E[T] = E[X] \cdot \lambda$$

Expected time found by  
 multiplying  $E[X]$  by the  
 time between each person